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Improving Uncertainty Estimation with Confidence-aware Training Data

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Abstract

AI-driven second-opinion systems play a crucial role in decision-making, especially in medicine, where accurate predictions guide clinicians. However, quantifying uncertainty in deep learning is challenging, as current methods often rely on hard class labels, which do not reflect true prediction confidence. This often results in overconfident predictions and slow convergence to true probabilities.

To address this, we suggest a new method that separates uncertainty into two types: epistemic and aleatoric. We estimate these uncertainties using hard and soft confidence labels, with experts providing confidence levels that indicate the likelihood of misclassification. We release an updated blood typing dataset consisting of 3139 images with soft labels of uncertainty annotations from six experts and hard labels collected from medical records. Proposed approach improves SotA uncertainty estimation quality by two times for blood typing (classification) and by 62% for histology (segmentation)[1].

1. Introduction

Domain experts often rely on AI-driven second-opinion systems backed by deep learning models to help make a decision or highlight additional areas in an image to explore [10]. While they are widely used, it is critical — especially in medicine — to quantify the confidence of a model in the provided predictions [2, 26]. They can also help identify erroneous or corrupted samples, guiding a laboratory assistant on what tests require a new trial. Uncertainty estimation (UE) for deep learning models [29] is common in natural language processing (NLP) and computer vision (CV) [1]. Thus, it can be incorporated into the

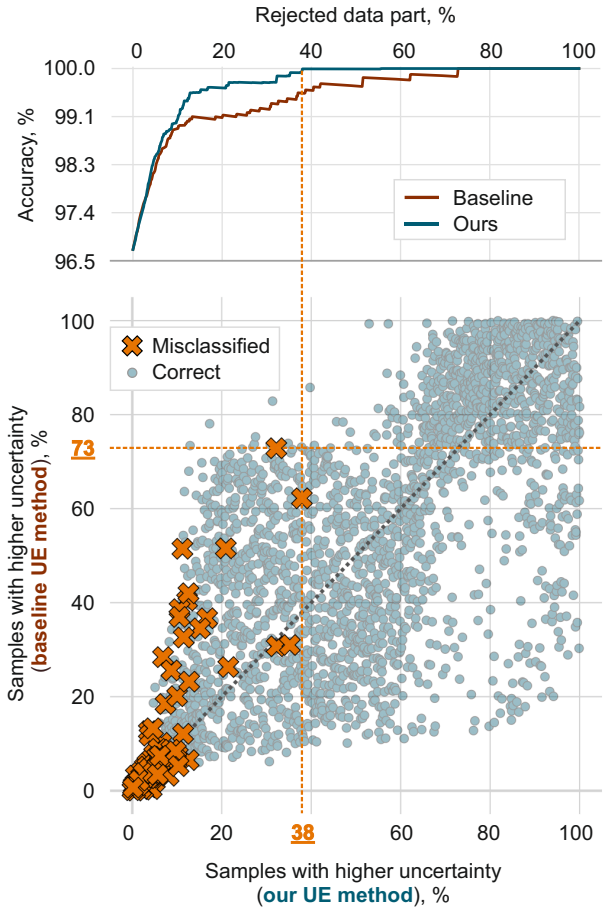


Figure 1. The proposed and baseline UE methods on the classification dataset. The top plot displays the dependence of accuracy on the test data remaining after the rejection of the most uncertain samples. The bottom plot depicts the joint distribution of samples according to the part of data with higher uncertainty. The proposed method allows for detection of all errors two times better.

[1]The code is available at: https://github.com/createcolor/confidence-aware-uncertainty.

second-opinion systems to improve their reliability and accuracy [34].

Existing methods for UE [19] made significant progress in recent years. Assessing a prediction's uncertainty involves separately estimating its two components: *epistemic* (model) and *aleatoric* (data) uncertainty [53]. Numerous methods exist for estimating epistemic uncertainty based on Bayesian ensembles and feature space analysis [19]. However, when addressing aleatoric uncertainty, the results often exhibit bias and demonstrate a slow convergence to the true uncertainty values, requiring thousands of images for confident uncertainty estimates [47]. In general this leads to a gap between true uncertainty values and their estimation [8,32], due to the lack of soft labels in the training data: we have access only to hard class labels without a degree of certainty about these labels. In this almost unsupervised case, a model produces overconfident predictions with slow convergence to true probabilities, as empirical and theoretical results suggest [7].

A potential solution to the mentioned problem is to use soft labels from experts, where instead of binary labels, the expert provides a confidence level indicating the probability of misclassification [6]. This transforms a weakly supervised problem into a fully supervised one, confidence-aware UE. However, relying solely on expert uncertainty is insufficient, as experts and neural networks often consider different factors. Experts may struggle to assess model uncertainty because they focus on domain specifics rather than deep learning capabilities. Additionally, expert labels often contain noise, particularly in classification [22] and segmentation tasks [28, 46].

Considering these observations, we propose a method that estimates both aleatoric and epistemic uncertainty. It utilizes soft expert confidence labels and hard binary ground truth labels to produce superior uncertainty estimates as shown on Fig. 1. By incorporating soft expert labels, we can more accurately model aleatoric uncertainty, while using binary labels ensures that the model remains grounded in the underlying problem's objective reality. Our main **contributions** are the following:

- We propose a new perspective on uncertainty quantification as a combination of epistemic uncertainty obtained with binary labels and aleatoric uncertainty obtained with additional soft confidence labels.

- Our experiments on synthetic data show that a supplement of a dataset with a few soft labels can significantly improve the model's UE degree. For example, if there are less than 2000 binary labels, additional 125 soft labels increase quality by 20%.

- Using real datasets, we demonstrated that the proposed method allows us to achieve a twofold improvement for the classification task and a 62% improvement with true expert uncertainties on the segmentation task compared to the SotA UE methods.

- We release a classification dataset for blood typing of 3139 with uncertainty markings by six experts in addition to available binary labels[2].

2. Literature review

Given the importance of UE in neural networks, a substantial body of research has emerged in this area [19].

On the one hand uncertainty can be predicted with a single deterministic network which gives the prediction based on one single forward pass. The primary benefit of this category lies in the straightforward nature of the problem statement: we can choose to view network confidences as a way to quantify uncertainty [39,45] or to train a distinct network for this purpose [41,42].

A more principled way to UE is the Bayesian treatment which covers all kinds of stochastic DNNs and posterior analysis of several different deterministic networks at inference. It provides a probabilistic interpretation of the model's outputs, replacing point estimates with distributions or confidence intervals. So, the variance of a predictive distribution reflects the uncertainty of the prediction [20]. While being introduced for neural networks quite early [13], this nice idea comes with its own challenges. Exact distributions for neural network predictions are analytically intractable, so one relies on approximations based on e.g. Variational inference [24] or Monte Carlo dropout [18].

A successful approximation of Bayesian techniques involves training an ensemble of models to treat discrepancies in their outputs as uncertainty estimates [35]. Neural networks are made as variable as possible so that they represent different feature spaces. The diversity comes from using different neural network architectures [23], training with different data subsamples [9], or using different augmentations. While being rather accurate, this approach has limited capacity to match the true distribution [52] and requires runs of many models instead of one.

Other required note is that we need to estimate both aleatoric and epistemic uncertainties to get a generally accurate values [11, 14, 48]. For regression, variance-based approaches are common [4], while classification often uses entropy-based measures [39]. However, entropy measures have limitations [53]. Variance-based methods, as shown in recent studies, address these issues and offer detailed uncertainty estimates at both the sample and label levels [15,44]. So, with Bayesian techniques, we can succeed in modeling both components.

[2]The dataset is available at: https://zenodo.org/records/14285035.

Uncertainties can be combined using a more straightforward approach based on ranking them against an independent dataset, as described in the work [49]. This method enables the consideration of uncertainties with varying magnitudes, although it may lack a robust theoretical foundation.

While the techniques mentioned above provide reasonable estimates of epistemic uncertainty, the inherent nature of aleatoric uncertainty makes it impossible to perfectly align with ground truth values without incorporating examples of uncertainty during training. A common approach to address this challenge is to use the diversity of expert opinions as a proxy for ground truth uncertainty. For instance, in [41], the authors model expert predictions as a form of uncertainty, while [12] leverages expert uncertainty feedback to guide the evolution of a neural network. Expert evaluations are also employed to validate the predicted uncertainty in [6]. Additionally, in [51], a hybrid system is introduced, where the final classification is chosen by selecting the most appropriate value between the neural network's output and expert predictions. While efficient, these approaches rely solely on expert labels, ignoring the existence of other possible labels that are typically available.

Despite the volume of research in UE for neural networks, most existing methods tend to produce overconfident predictions due to their reliance on hard labels rather than incorporating soft estimations of confidence for objects. Approaches that take into account expert-based uncertainty are rare, and even fewer studies address scenarios where both expert labels and ground truth labels coexist. However, there are a sufficient number of medical datasets [3, 5, 33, 36, 37, 40] annotated by multiple experts, where any of the experts markup can be used as a binary label, and their combination as a soft label. This gap highlights the need for a more advanced UE framework designed for neural networks, one that can seamlessly integrate both sources of labeling to achieve more accurate and reliable uncertainty estimation. Developing such a framework would represent a critical advancement in the field, offering substantial improvements in model confidence and decision-making reliability, particularly in high-stakes applications, including CV for medicine.

3. Uncertainty estimation with confidence-aware training data

This section presents a discussion of estimating uncertainty theoretical aspects. For simplicity we consider binary classification tasks. We focus on decomposing total variance into epistemic and aleatoric components and on a novel procedure for estimating these components.

3.1. Uncertainty decomposition

Let $y \in \{0, 1\}$ represent the class label and x the input image. We model the class label y as a binary random variable following a Bernoulli distribution given x, denoted $y|x \sim \text{Bern}(p(x))$, where $p(x) = \mathbb{P}(y = 1 \mid x)$.

The aim of a model $p_\theta(x)$ with parameters θ is to predict the probability of an object x to have class $y = 1$: $p_\theta(x) = \mathbb{P}(y = 1 \mid x, \theta) \approx \mathbb{P}(y = 1 \mid x)$. Consequently, the predicted expected value and the variance of y given x and θ are:

$$\mathbb{E}(y \mid x, \theta) = p_\theta(x), \mathbb{V}(y \mid x, \theta) = p_\theta(x)(1 - p_\theta(x)). \quad (1)$$

The uncertainty in predicting the class label y according to law of total variance [38] can be decomposed as follows:

$$TV \stackrel{\text{def.}}{=} \mathbb{V}(y \mid x, \mathcal{D}) = \underbrace{\mathbb{V}_{\theta \sim \mathbb{P}(\theta|\mathcal{D})} \left[\mathbb{E}(y \mid x, \theta)\right]}_{\text{epistemic}} + \underbrace{+ \mathbb{E}_{\theta \sim \mathbb{P}(\theta|\mathcal{D})} \left[\mathbb{V}(y \mid x, \theta)\right]}_{\text{aleatoric}}. \quad (2)$$

The first term represents epistemic uncertainty, which arises due to the model's incomplete knowledge about the true data distribution due to limited training data. This uncertainty reflects how much predictions can differ across various models or parameter sets sampled from $\mathbb{P}(\theta \mid \mathcal{D})$. Epistemic uncertainty is reducible with more data and, mathematically, is captured as the variance in predictions across different values of θ.

In contrast, aleatoric uncertainty from the second term is inherent in the data and irreducible, even with additional data. It stems from factors like noise, ambiguity, or inherent randomness given the input x. This uncertainty is represented mathematically as the expected variance predicted by models, reflecting the ambiguity present in the data.

In practice, these expressions are approximated via Monte Carlo methods:

$$\mathbb{V}_\theta \left[\mathbb{E}(y \mid x, \theta)\right] \approx \frac{1}{K} \sum_{i=1}^{K} \left(p_{\theta_i}(x) - \overline{p_\theta}(x)\right)^2, \quad (3)$$

$$\mathbb{E}_\theta \left[\mathbb{V}(y \mid x, \theta)\right] \approx \frac{1}{K} \sum_{i=1}^{K} p_{\theta_i}(x) \left(1 - p_{\theta_i}(x)\right), \quad (4)$$

where K is the number of trained models, θ_i are parameters of i-th model, $\overline{p_\theta}(x) = \frac{1}{K} \sum p_{\theta_i}(x)$ is the average prediction of the models.

3.2. Confidence-based aleatoric uncertainty model

While the epistemic uncertainty ensemble approach provides a reasonable uncertainty estimate, for aleatoric uncertainty in Eq. (2), we get poor estimates caused by the overconfidence of neural networks [21]. Therefore, improving aleatoric uncertainty estimates is essential for enhancing the overall quality of uncertainty evaluation. For that purpose, we might: 1) provide soft labels to overcome overconfidence and 2) achieve better estimation of the variance $\hat{p}(x)(1 - \hat{p}(x))$ having more precise $\hat{p}(x)$.

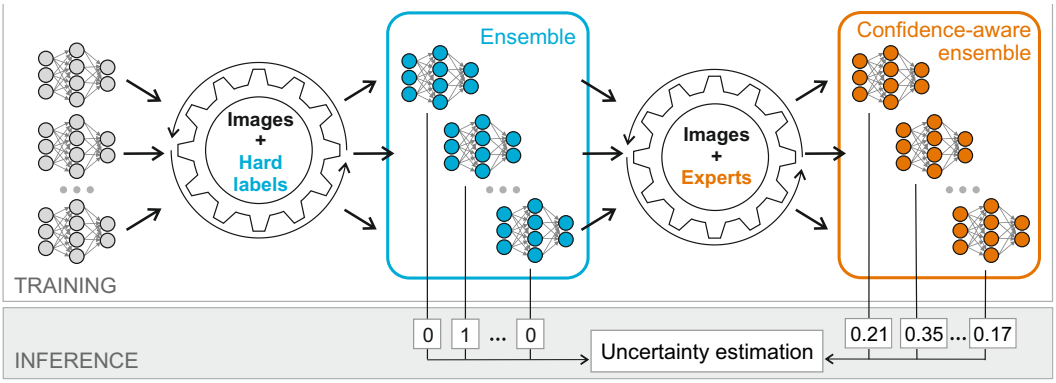


Figure 2. The proposed expert-aware uncertainty estimation training and inference pipeline.

The estimates of $p(x)$ provided by experts are available in the training sample. So, by learning to imitate them with a model $p_e(x)$, we obtain a more accurate $p_e(x)(1-p_e(x))$. This alternative approach can provide a more reliable measure of uncertainty in scenarios where standard models fail to capture the full extent of aleatoric uncertainty. We note that this approach would be methodologically incorrect for estimating epistemic uncertainty, as an expert can't qualify what a model is aware of.

3.3. Confidence-aware ensemble approach

Given the developed decomposition and the intuition above, we propose a straightforward general approach for uncertainty estimation based on the law of total variance. This approach requires data such that for a sample x, there can be available two types of markup: a hard label $y \in \{0, 1\}$ and a soft label $\tilde{y} \in [0, 1]$ associated with expert assessments. First, an ensemble of neural networks is trained to predict hard labels and then fine-tuned to predict soft labels.

The ensemble trained on hard labels approximates epistemic uncertainty as given by (3). On the other hand, using the same ensemble for aleatoric uncertainty evaluation as given by (4) may fail to provide sufficiently adequate results. Therefore, we use the ensemble trained on probability labels to generate p_{θ_i} values in (4).

Figure 2 presents the final pipeline for our method. Specifically, the training has the following steps: 1) train an ensemble of K models p_{θ_i} to predict the hard ground truth labels; 2) fine-tune these models using a set of expert labels to get another ensemble of K models $\tilde{p}_{\theta_i}$. The inference also consists of two steps: 1) use (3) to calculate the epistemic uncertainty with the ensemble $\{p_{\theta_i}\}_{i=1}^K$; 2) use (4) to calculate the aleatoric uncertainty with the ensemble $\{\tilde{p}_{\theta_i}\}_{i=1}^K$.

While other similar pipelines are possible, this one correlates most with our theoretical justifications and possible scarcity of expert labels in data — and provides superior or similar results to considered alternatives that use the same data samples.

4. Overconfidence suppressing hypothesis verification

Previous research has highlighted that in high-dimensional settings, models often exhibit overconfidence and poor uncertainty estimation quality [47]. Particularly calibration refers to the alignment between predicted probabilities and actual outcomes; a well-calibrated model will output probabilities that accurately reflect the true likelihood of an event occurring ($p_\theta(x) \approx p(x)$). We hypothesize that replacing hard binary labels with soft probabilistic targets, such as estimated probabilities, during training could improve model calibration by allowing the model to better capture uncertainty. We conducted numerical experiments for artificial data initiating from the methodology outlined in [47] to test this hypothesis.

4.1. Experimental setup

In experiments, we generated synthetic data by sampling input features independently from a Gaussian distribution, $x \sim \mathcal{N}\left(0, \sigma_x^2 I_d\right)$, with $\sigma_x = 2.5 \cdot 10^{-4}$ and dimensionality $d = 800$. The coefficients β were also sampled independently, following $\beta_i \sim \mathcal{N}(3, 16), i \in \overline{1, 800}$. These specific parameters ensured that the magnitude of the log-odds ratio remains controlled, preventing the logistic function from collapsing to trivial values (i.e. probabilities close to 0 or 1) that could arise for large absolute values of the log-odds.

The binary target variable y comes from a Bernoulli distribution, where the logistic function determines the success probability:

$$p(x) := \mathbb{P}(y = 1 \mid x, \beta) = \frac{1}{1 + e^{-x^\top \beta}}. \tag{5}$$

To simulate noisy or rounded estimates of the true probabilities and investigate their impact on model calibration, we

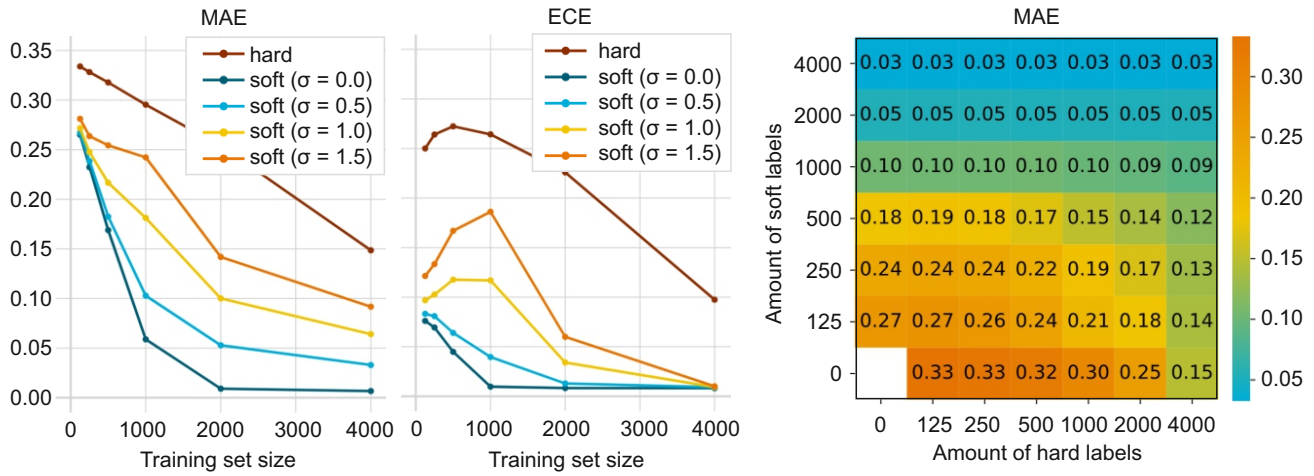


Figure 3. The first two plots demonstrate how MAE and ECE vary with changes in training set size and the usage of hard/soft/corrupted soft labels. The third one highlights MAE values in a scenario with initial training on hard labels and fine-tuning on soft labels.

added Gaussian noise to the logits during training:

$$\tilde{y} = \frac{1}{1 + e^{-x^\top \beta + z}}, \; z \sim \mathcal{N}(0, \sigma_z). \tag{6}$$

This data generation scheme allowed us to compare the performance and calibration of high-dimensional logistic regression models trained with binary targets (hard labels) versus those trained with noisy probabilistic targets (soft labels). The experiments involved training high-dimensional logistic regression models with varying training set sizes $N_{train} \in \{125, 250, 500, 1000, 2000, 4000\}$. For models trained with probabilistic soft labels, we employ different noise levels $\sigma_z \in \{0.0, 0.5, 1.0, 1.5\}$. All models were trained using standard gradient-based methods, minimizing cross-entropy loss.

For evaluation, we used a test set of $N_{test} = 10000$ samples to calculate the Mean Absolute Error (MAE) and Expected Calibration Error (ECE) [21]. For the ECE, we used $M = 10$ bins. MAE measures the average absolute difference between the predicted and the true probabilities, which are available in this synthetic setting. In contrast, ECE groups predictions by confidence levels and calculates the weighted average of the absolute differences between predicted confidences and actual accuracies within each group.

Additionally, we conducted experiments to evaluate the impact of combining hard and soft labels, using $\sigma_z = 0.5$ for the latter. We expected that pre-training on a substantial amount of hard labels would help the model capture essential patterns, while fine-tuning on a smaller set of soft labels would improve the model's calibration and its ability to estimate probabilities accurately. This setup simulated real-world scenarios where binary labels are more accessible than uncertainty-aware confidence labels.

4.2. Results

The relationship between model performance and training size is shown in Fig. 3. The first two plots illustrate how models trained on different types of labels perform in terms of MAE and ECE, and the heatmap reveals patterns when both label types are combined during training.

The decrease in MAE with increasing training size is expected. ECE initially increases with training size because for a small dataset the model learns very little and behaves randomly, resulting in predictions that cluster around 0.5. This aligns with accuracy, leading to a low ECE. Figures show that even with noisy probability estimates, models trained on soft labels perform better than those trained on binary labels, demonstrating the effectiveness and robustness of soft targets. Additionally, models trained on soft labels reach lower error rates more quickly as the training size increases. For example, with a moderate level of noise ($\sigma_z = 1.0$), training with only 250 samples achieves comparable MAE performance to training on 2000 hard labels.

The heatmap in Fig. 3 illustrates the MAE of models trained with varying combinations of hard and soft labels for pre-training and fine-tuning respectively. Interestingly, we observe that even with a relatively small number of soft labels, fine-tuning after pre-training on hard labels leads to significant performance improvements. For instance, using 2000 hard labels and 125 soft labels together achieves a synergistic effect, reducing MAE by approximately 33% compared to using only soft labels, and by 28% compared to using only hard labels. This result highlights the practicality of a combined approach in real-world settings, where confidence labels may be expensive or difficult to collect in large quantities.

Therefore, experts are crucial because they provide soft

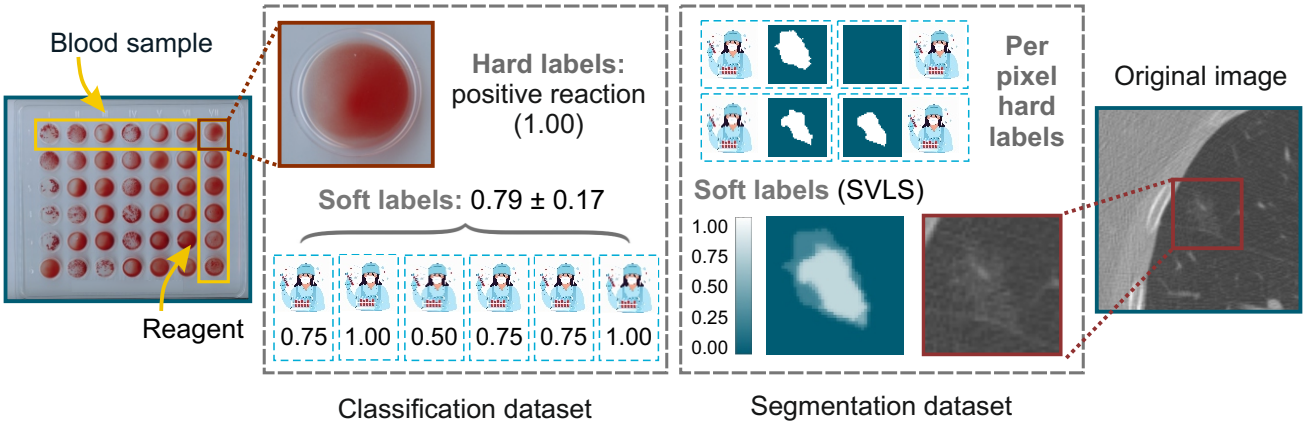


Figure 4. Structure of datasets utilized in the work. In the classification dataset (left part), each sample is annotated with a binary label and six soft labels from experts. In the segmentation one (right part), each sample has four binary maps and a soft labels SVLS map.

probability estimates that improve upon those from high-dimensional models, leading to better capture of aleatoric uncertainty. Additionally, incorporating even a small number of soft labels into the training process can significantly enhance model performance by reducing overconfidence and improving calibration, which are essential for accurate estimation of aleatoric uncertainty.

5. CAE for classification and segmentation

To investigate suitability of our method for real-life problems, we study two different computer vision tasks: image classification and semantic segmentation. In accordance with Eq. (2) we denote by $TV(\mathrm{X}, \mathrm{Y})$ a method that evaluates uncertainty by combining epistemic uncertainty from predictions X (computed via Eq. (3)) and aleatoric uncertainty from predictions Y (computed via Eq. (4)). $TV(\mathrm{X}, -)$ implies that only epistemic uncertainty is used for evaluation, and vice versa for $TV(-, \mathrm{Y})$ and aleatoric uncertainty.

In our experiments, we utilized three sources of predictions: an ensemble trained on hard labels (further denoted by Ens), a Confidence-Aware Ensemble trained on soft labels (CAE), and expert assessments (Exps). Combining these sources in different ways, we obtain $TV(\mathrm{Ens}, -)$, $TV(\mathrm{Ens}, \mathrm{Ens})$, $TV(\mathrm{Ens}, \mathrm{Exps})$ and $TV(\mathrm{Ens}, \mathrm{CAE})$. The third method requires expert assessments during inference which is often not applicable in real life. The last method relies on expert assessments only during training and not inference. We consider it to be our main method.

We additionally study the applicability of $TV(-, \mathrm{Exps})$. This method, however, performs poorly in segmentation tasks where it practically has no ability to locate pixel-level errors produced by a model if experts provide consistent evaluations on those pixels. Therefore we only evaluate $TV(-, \mathrm{Exps})$ in the classification task.

The experimental analysis investigates the applicability of our method by comparing it to diverse baselines including SotA methods. For the classification problem we consider MCMC [50], MC Dropout [18] and HUQ [49]. For the segmentation problem we analyze results from MCMC and ABNN [17] methods.

5.1. Datasets with expert annotation

To demonstrate the full capabilities of our method, we chose two datasets which provide sample assessments by several medical experts. The first considers a classification problem, and the second one has segmentation labels and targets. They both have similar structures depicted at Fig. 4.

For image classification, we equipped the BloodyWell dataset with expert soft labels [31]. This dataset contains images of mixtures of blood samples and various reagents for blood typing. The task is to classify images based on whether the agglutination reaction had occurred in the mixture. In total, there are 3139 images where each image is annotated with a binary label based on the patient's medical history and independent assessments from 6 qualified serology experts. Experts were asked to assign each image a fixed probability from $\{0.0, 0.25, 0.5, 0.75, 1.0\}$ based on their certainty that a reaction had occurred. While this study incorporates uncertainties from all six experts, we also performed an experiment to demonstrate that a carefully selected combination of three experts yields nearly identical results, as presented in the supplementary material.

For semantic segmentation, we utilized the LIDC-IDRI dataset [5]. It contains 1018 lung CT scans annotated by 4 radiologists for nodule segmentation with binary maps. Consistency of their responses in each pixel is considered to be average confidence. In our experiments, volumetric scans were divided into 2D slices, which were then cropped

to images of size 128×128 centered around the nodule annotation. In total, 15096 images were acquired.

5.2. Experimental setup

General setup. We trained an ensemble of $K = 10$ models for classification and segmentation. Each model employs different train/validation splits. During training, Adam optimizer [30] with $\beta_1 = 0.9$, $\beta_2 = 0.999$ was used for parameter optimization. Cross-entropy was chosen as a loss function for both classification and segmentation.

Classification. Extended BloodyWell dataset was split in proportion of 1:1:8 for training, validation and testing, respectively (this helps to balance easy and hard examples calculated with expert consensus). The classification ensemble was trained using hard labels. Then, for each image, we calculated a mean expert vote given by $\tilde{y} = \frac{1}{6}\sum_{e=1}^{6} y_e$, where y_e is individual expert's confidence assessments. These values then were used to fine-tune the models.

We chose MobileNet-V3-small [25] as a simple and reliable classifier architecture. To account for the peculiarities of the dataset we added information about the used reagent through one-hot encoding at one of the neural network's dense layers. Initially, the classification ensemble was trained for 400 epochs with a scheduled decreasing learning rate of $4 \cdot 10^{-4}$ and a weight decay of $5 \cdot 10^{-5}$. Then, the ensemble was fine-tuned to predict mean expert votes for 300 epochs. The training process brings variety to produce different feature spaces for each individual network through standard data augmentations, different initializations, a dropout layer, and random splits between train and validation sets.

Segmentation. The LIDC-IDRI dataset used for segmentation naturally does not have ground truth labels. For that reason, we conducted four experiments, each using a different expert as the ground truth. The data was split in ratio of 6:2:2 for training, validation, and testing, respectively.

To combine expert assessments, we adopted multi-rater spatially varying label smoothing (mSVLS) [27]. Specifically, each expert's assessment was smoothed via convolution with a 3×3 weight matrix W:

$$\tilde{y}_{e,(x,y)} = \frac{1}{\sum W_{(i,j)}} \sum_{i=1}^{3}\sum_{j=1}^{3} y_{e,(x+i-2,y+j-2)} W_{(i,j)}, \quad (7)$$

where $y_{e,(x,y)}$ is expert's prediction for pixel (x, y). Then these smoothed maps were averaged by expert to get $\tilde{y}_{(x,y)} = \frac{1}{4}\sum_{e=1}^{4} \tilde{y}_{e,(x,y)}$. This approach to training is shown to produce better-calibrated models.

As an architecture for segmentation, we chose a classical UNet [43]. An ensemble was trained on hard labels y_i and then fine-tuned on soft labels $\tilde{y}_i$, both for 100 epochs with a learning rate of 10^{-4} and a weight decay of 10^{-4}.

Detailed descriptions on how we trained baseline solutions can be found in the supplementary material.

5.3. Uncertainty estimation metrics

A clear and understandable approach to evaluating and comparing UE techniques is through the use of rejection curves. To plot a rejection curve, one first chooses a throwaway rate t between 0% and 100%. A fraction t of samples with the highest uncertainty values is then thrown away, and the network is evaluated on the remaining samples. A rejection curve demonstrates quality metric values a network achieves against various throwaway rates.

For classification task, we rejected images with high uncertainty entirely and used accuracy to evaluate model performance. For segmentation, we discarded a fraction t of pixels with the highest uncertainty on each image and evaluated the model via mean Dice score.

Despite their usefulness, rejection curves don't allow for direct quantitative comparison of methods. For that reason, we compute the Area Above the rejection Curve (AAC), normalized via subtracting the area above the oracle curve that rejects all mistake samples at first, following [16]. An ideal uncertainty measure would produce the steepest rejection curve and the smallest AAC. We also compare TRA-99, the Throwaway Rate required to attain a classification Accuracy of 99%, and TRD-95, the Throwaway Rate required to attain a segmentation Dice score of 95%.

5.4. Results

Ten neural networks of a *classification* ensemble achieved an accuracy of 0.9671 ± 0.0058 on the test subset. We first calculated rejection curves for each neural network and then averaged them to guarantee the stability of results for target UE methods. Averaged rejection curves are presented in Fig. 5 while their detailed statistics are in Tab. 1.

For more than 80% of the classification dataset all six experts are actually in complete agreement with each other which relates to our setup on synthetic dataset with mixed amounts of hard and soft labels in Sec. 4. The mean ECE of the trained ensemble is 0.0245, while the ECE of the experts' estimate is 0.0217, which is 12% lower. Interestingly, a common calibration method — calibrator training — failed to improve calibration of classification models. We elaborate on this in the supplementary material.

First of all, it is worth mentioning that using Total Variance shows significant improvements compared to methods that exclusively assess epistemic uncertainty. Adding an aleatory component to the classification ensemble leads to a 14% decrease in terms of AAC. Moreover, using either expert assessments or CAE as measures of aleatoric uncertainty improves metrics significantly, with the latter providing best results and improving AAC **twofold** compared to Total Variance of ensembles.

Models in the *segmentation* ensembles achieved an average Dice score of 0.6348 ± 0.0229 for four expert markings. Rejection curves for the segmentation task are shown

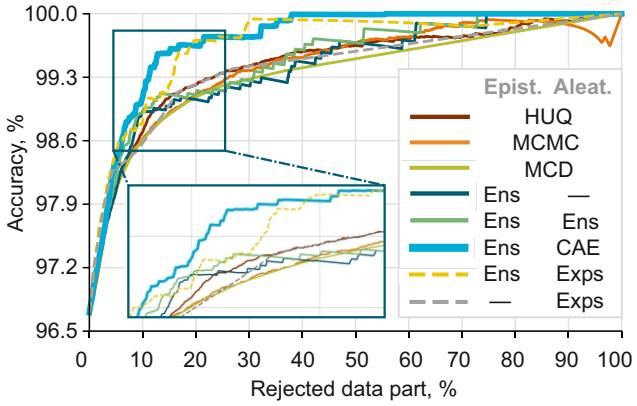


Figure 5. Accuracy-rejection curves for classification.

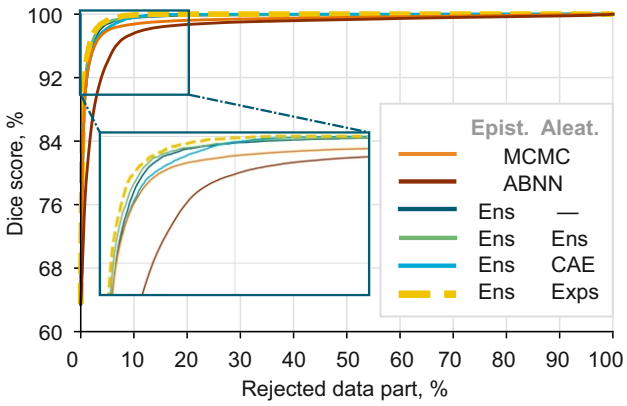


Figure 6. Averaged rejection curves for semantic segmentation.

UE method		AAC, $\times 10^{-4}$, ↓	TRA-99, %, ↓
Epist.	Aleat.		
MCMC [50]		51.7	17.7
MC Dropout [18]		59.1	17.7
HUQ [49]		49.7	15.0
Ens	–	50.0	15.4
Ens	Ens	43.6	11.8
Ens	CAE	**21.4**	**9.2**
–	Exps	51.2	15.1
Ens	Exps	**27.5**	**10.7**

Table 1. Quantitative comparison of uncertainty estimation methods for image classification. The two best results are in bold, and the best result is underlined.

UE method		AAC, $\times 10^{-4}$, ↓	TRD-95, %, ↓
Epist.	Aleat.		
MCMC [50]		68.5	2.06
ABNN [17]		145.1	5.73
Ens	–	29.4	1.86
Ens	Ens	**24.3**	**1.56**
Ens	CAE	27.7	2.03
Ens	Exps	**15.0**	**1.30**

Table 2. Quantitative comparison of uncertainty estimation methods for semantic segmentation, averaged over four experimental setups. The two best results are in bold, and the best result is underlined.

in Fig. 6 and statistics are presented in Tab. 2, averaged over four experts. Full results for individual experts can be found in the supplementary material. Notably, methods that combine epistemic and aleatoric uncertainty still outperform other methods with only one exception. For example, a combination of uncertainties for regular ensembles produces AAC from 9% to 16% lower than just epistemic uncertainty with a similar tendency for TRD-95.

In contrast to classification, using CAE for aleatoric uncertainty estimation outperforms on regular ensembles for only one expert, while producing comparable or slightly worse results in other setups. We theorize that this is due to the relatively low expert agreement on LIDC: although consensus was achieved during the initial labeling process, numerous instances exist where labels from one or multiple experts are absent. This potentially negatively impacts CAE performance. To support this claim, additional experiments were conducted on the RIGA dataset with better expert agreement. Results are presented in the supplementary material.

6. Conclusion

In this work, we proposed a novel method for uncertainty quantification that effectively captures both aleatoric and epistemic uncertainty by leveraging the law of total variance. This method integrates traditional ensemble-based uncertainty with ensemble estimates trained with soft labels provided by experts, offering a more nuanced and reliable measure of model confidence. Our approach addresses the common overconfidence issue in deep learning models by incorporating expert feedback, which helps align model predictions with true uncertainties.

We provided mathematical intuition and validated the effectiveness of our method through experiments in classification and segmentation tasks, demonstrating twofold and 67% improvement respectively. To support further research, we release an extension to blood-typing dataset, new version has 3139 images annotated by binary labels and six experts‘ confidence values.

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